

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Department of Metallurgical and Materials Engineering

2. Subject Name: Advanced Materials
Credits: 3

L-T-P: 3-0-0

3. Pre-requisites: None

4. Syllabus and reference books:

Syllabus:

- Nanomaterials and thin films: Classification, Properties, Synthesis, Bulk nanomaterials, Processing and Applications
- Bulk metallic glasses, High entropy alloys and ceramics: Fundamentals, synthesis / processing, properties and applications
- High temperature materials: Metallic - superalloys and intermetallics and ultra high temperature ceramics and composites
- Cellular materials: Metallic foams and porous ceramics
- Energy Materials: Fuel cells, Battery, Super-capacitors, Storage materials, Thermoelectric materials
- Functional materials: Dielectric, Piezo, Pyro, Optoelectrical and Magnetic materials: Fabrication and applications
- Semiconducting materials
- 2D materials: CNT, Graphene, hBN, Chalcogenides, Tellurides

Reference Books:

Semiconductor Fabrication and design:

1. Senturia, Stephen D. Microsystem design. Springer Science & Business Media, 2005.

Semiconductor Physics:

1. Wenckebach, W. Tom. Essentials of semiconductor physics. John Wiley & Sons, 1999.

2. Martins, Emiliano R. Essentials of semiconductor device physics. John Wiley & Sons, 2022.

3. Nanostructured Materials: Processing, Properties, and Applications, Ed. C.C. Koch, Nova Publications, IOP, p612

4. Bulk Metallic Glasses, by C. Suryanarayana, C. and A. Inoue, CRC Press, p543

5. High-Entropy Alloys: Fundamentals and Applications, Ed. M.C. Gao, J.W. Yeh, P.K. Liaw and Y. Zhang, Springer

6. Huggins, Robert. Advanced batteries: materials science aspects. Springer Science & Business Media, 2008

7. Ravindra, N. M., Jariwala, B., Bañobre, A., & Maske, A. (2018). Thermoelectrics: fundamentals, materials selection, properties, and performance. Springer

8. Lee, HoSung. "Thermoelectrics: Design and Materials." (2018): 712-713

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Nanomaterials and thin films	7
2.	Bulk metallic glasses, High entropy alloys and ceramics	6
3.	High temperature materials	5
4.	Cellular materials	3
5.	Energy Materials	5
6.	Functional materials	6
7.	Semiconducting materials	4
8.	2-D materials	4
Total number of hours		40